ADCS Final Project: Tracking *El Niño*

Master's Degree in Aerospace Engineering

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1 Introduction

The purpose of this project is to design and implement a control algorithm capable of maintaining a satellite's pointing direction towards a specific geographic point on Earth throughout its orbital motion. The selected point of interest (POI) lies near the equatorial Pacific Ocean, where the El Niño phenomenon occurs.

This exercise represents an application of attitude dynamics and control principles using realistic orbital and physical parameters. The project's structure follows the given prompt, divided into two main sections: (1) orbital geometry and attitude representation, and (2) attitude dynamics and control.

All computations, numerical integrations, and visualizations were implemented in MATLAB, including custom functions to handle quaternion algebra, rotation matrices, and animation of the satellite's attitude.

2 Mission Scenario and Parameters

The satellite orbits the Earth in a low circular orbit at an altitude of approximately 700 km. The Earth's rotation is included with an angular velocity $\omega_E = 2\pi/86400\,\text{rad/s}$. The main constants and parameters used in the simulation are summarized in Table 1.

Table 1: Main constants and orbital parameters used in the simulation.

Parameter	**Symbol / Expression**	**Value**
Gravitational constant	G	$6.6743 \times 10^{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$
Earth mass	M_E	$5.972 \times 10^{24}\,\text{kg}$
Earth radius	R_E	$6.371 \times 10^{6}\,\text{m}$
Gravitational parameter	$\mu_E = GM_E$	$3.986 \times 10^{14}\,\text{m}^3/\text{s}^2$
Earth rotation rate	ω_E	$7.292 \times 10^{-5}\,\text{rad/s}$
Orbit altitude	h	$700 \times 10^{3}\,\text{m}$
Orbit radius	$r_{\text{orbit}} = R_E + h$	$7.071 \times 10^{6}\,\text{m}$
Orbital period	$T_{\text{orbit}} = 2\pi\sqrt{r_{\text{orbit}}^3/\mu_E}$	$\approx 5940\,\text{s}$ ($\sim 99\,\text{min}$)

The point of interest (POI) is defined by its latitude and longitude (latitude = 0°, longitude = –78.47°). The simulation time covers one orbital period, discretized into several hundred time steps for numerical integration.

3 Section 1: Satellite Geometry and Orbital Representation

The geometry of the studied satellite is the following:

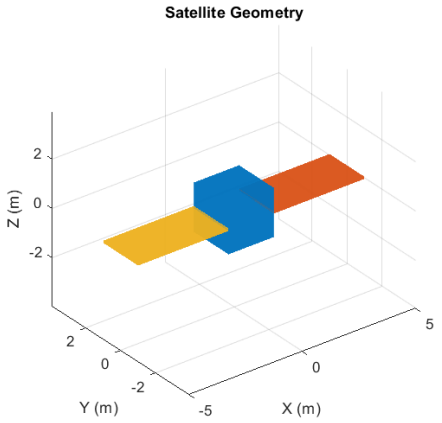


Figure 1: Geometry of the satellite

With an inertia tensor of (in I.S. units):

$$A = 1.0 \times 10^3 \begin{bmatrix} 0.3001 & 0 & 0 \\ 0 & 1.3001 & 0 \\ 0 & 0 & 1.3333 \end{bmatrix}$$

3.1 Earth and Orbit Representation

A 3D representation of the Earth was created, displaying coastlines and the equatorial plane. The satellite's orbit was defined as a circular path with constant radius

$$r_{\text{orbit}} = R_E + h \tag{1}$$

and both the Earth's rotation and the satellite's orbital motion were expressed in the Earth-Centered Inertial (ECI) frame.

3.2 Satellite and Point of Interest Position

At each simulation step, the ECI position of the satellite $\mathbf{r}_{\text{sat}}(t)$ and the rotating position of the POI $\mathbf{r}_{\text{POI}}(t)$ were computed. The line-of-sight vector from the satellite to the POI was then given by:

$$\mathbf{r}_{\text{POI/sat}} = \mathbf{r}_{\text{POI}} - \mathbf{r}_{\text{sat}} \tag{2}$$

which was normalized to obtain the desired pointing direction for the control system.

3.3 Satellite Body Reference Frame (BRF)

The Body Reference Frame (BRF) of the satellite was introduced to represent its instantaneous attitude with respect to the inertial frame. In the MATLAB animation, the BRF was visualized as three orthogonal axes rigidly attached to the satellite, rotating according to its angular velocity vector $\boldsymbol{\omega}_{\text{sat}}$.

The position of any point in the body frame, $\mathbf{r}_b$, was transformed to the inertial frame using the attitude rotation matrix R_{bi}.

This allowed the 3D representation to update in real time, showing the rotation of the satellite axes as the attitude evolved under the influence of the control torques.

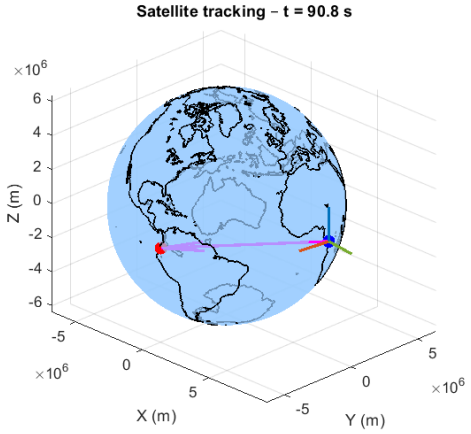


Figure 2: Geometry of the problem representation. Frame of the animation

4 Section 2: Attitude Dynamics and Control

4.1 Desired Quaternion

At each instant, the desired orientation quaternion q_d was computed so that the satellite's body Z-axis (sensor axis) pointed directly toward the POI. This was achieved by constructing a rotation that aligns the current pointing vector $\mathbf{b}_{\text{sensor}}$ in the body frame with the target direction $\mathbf{v}_{\text{des}}$ in the inertial frame:

$$\mathbf{v}_{\text{des}} = \frac{\mathbf{r}_{\text{POI}} - \mathbf{r}_{\text{sat}}}{\|\mathbf{r}_{\text{POI}} - \mathbf{r}_{\text{sat}}\|} \tag{3}$$

4.2 Attitude Error Quaternion

The attitude error between the desired and actual orientations was expressed as:

$$q_e = q_d \otimes q^{-1} \tag{4}$$

where $\otimes$ denotes quaternion multiplication. The corresponding axis–angle representation was used to compute the physical angular error vector:

$$\mathbf{e}_{\text{ang}} = \hat{\mathbf{u}}\,\theta \tag{5}$$

where $\hat{\mathbf{u}}$ is the unit rotation axis and θ the rotation angle in radians. This form provides a linear measure of orientation error, better suited for control purposes than the raw quaternion vector part.

4.3 Rigid Body Dynamics

The rotational motion follows Euler's equation for a rigid body:

$$\mathbf{I}\,\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\text{ctrl}} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) \tag{6}$$

where $\mathbf{I}$ is the inertia tensor and $\boldsymbol{\tau}_{\text{ctrl}}$ is the control torque generated by the attitude controller. The satellite's quaternion rate equation was integrated simultaneously:

$$\dot{q} = \frac{1}{2}\,\Omega(\boldsymbol{\omega})\,q \tag{7}$$

with $\Omega(\boldsymbol{\omega})$ the skew-symmetric matrix representing the angular velocity.

4.4 PID Control

A proportional–integral-derivative controller was implemented in the body frame:

$$\boldsymbol{\tau}_{\text{ctrl}} = K_p\,\mathbf{e}_{\text{ang}} + K_i \int \mathbf{e}_{\text{ang}}\,dt + K_d\,(-\boldsymbol{\omega}) \tag{8}$$

The term $-\boldsymbol{\omega}$ provides damping to suppress oscillations. The integration was performed using an explicit Euler scheme over the simulation time vector.

Through numerical experimentation, several combinations of the control gains K_p, K_i, and K_d were tested to achieve stable and responsive pointing behavior.

5 Results and Discussion

The simulation produced smooth quaternion evolution and convergent angular velocity profiles. The proportional and derivative gains (K_p and K_d) dominated the control behavior, ensuring fast convergence of the attitude error without overshooting.

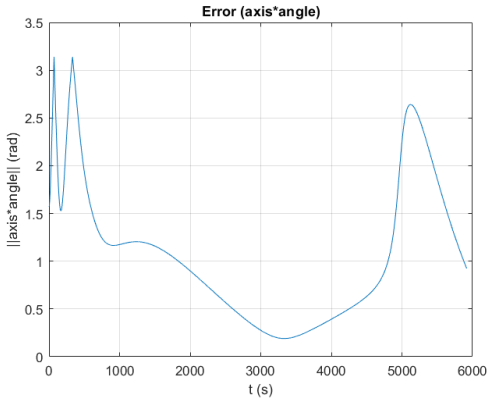
It was observed that the integral term (K_i) had to remain very small or even zero to maintain numerical stability. This is consistent with the nature of attitude dynamics: the quaternion kinematics inherently integrate angular velocity over time, thus already providing a form of integral action. Introducing an additional integral gain can lead to redundancy and possible drift in the control torque.

In the final run, the gains are:

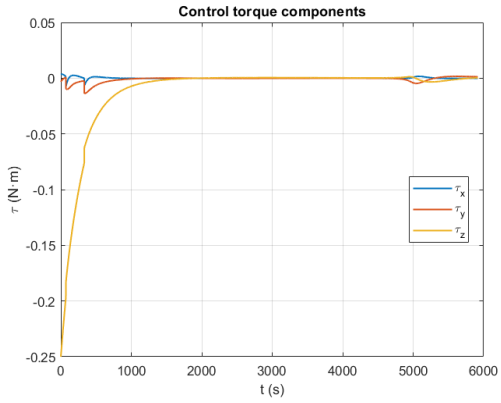
Table 2: Main constants and orbital parameters used in the simulation.

Parameter	**Value**
K_p	0.005
K_i	0
K_d	5

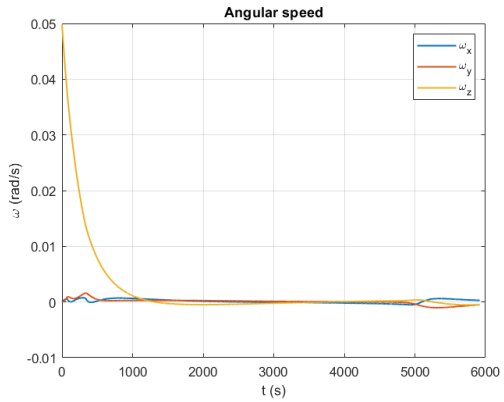
This control gains result in the following parameters during the studied orbit:



(a) Error magnitude.



(b) Control torques.



(c) Angular speed.

Figure 3: Results of the simulation

6 Conclusions

This project provided a comprehensive understanding of satellite attitude dynamics, quaternion-based attitude representation, and feedback control design. The developed MATLAB model successfully demonstrated the ability of a PDI controller to stabilize a satellite's pointing direction towards a ground target.

Two main insights were deduced from the numerical study:

1. Because the satellite's inertia tensor is not isotropic (the three principal moments of inertia differ), the ideal control law should employ **gain matrices $\mathbf{K_p}$** and $\mathbf{K_d}$ of size 3×3, instead of scalar constants. This would allow for axis-dependent tuning and compensation of the different rotational inertias.

2. The integral term's influence is secondary in this context, since the quaternion kinematics already provide an inherent integration of angular velocity. Therefore, a small or null K_i value is most effective, avoiding redundant accumulation and potential instability.

The project confirmed the theoretical foundations of quaternion-based control and highlighted practical considerations in controller tuning for real spacecraft systems. The experience also deepened the understanding of how orbital geometry, reference frames, and attitude dynamics interact in the context of Earth observation missions.